Machine Learning Analysis of T-Cell Receptor Sequences in Acute Respiratory Distress Syndrome

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ABSTRACT

ARDS, Acute Respiratory Distress Syndrome, is a lung condition known for the collection of edema protein fluid in the lung's air sacs which subjects patients into a hypoxic state. Within the US, approximately 190,600 patients are affected by ARDS every year with a mortality rate of ~45%. Recently, ARDS has been shown to occur secondarily during SARS-CoV-2 infections. Although severe ARDS is considered a fatal disease, people who survive ARDS and recover from SARS-CoV-2 may have lasting pulmonary scarring. The experiment hypothesis was that ARDS patients exhibit immune specific T-cell populations and that these sequences could be identified using high-throughput sequencing strategies. To do this, we employed cellular experiments using an Artificial Intelligence approach to quantify the viability of single cell fractions; this was accomplished using lung fluid taken from 42 ARDS patients. Next, we extracted genomic DNA from each of these samples followed by high-throughput sequencing using the Human T-cell receptor β (TCRβ) ImmunoSeq assay. Results from the cellular viability experiments showed that the range of viable cells across samples averaged between $1x10^6$ and $1x10^7$ cells per ul. Results from the immune sequencing experiments demonstrated that specific TCRβ genes could be tracked and utilized as markers to compare across samples. These genes could also be further quantified using their relative frequency which averaged from 4.12% to 33.98%. A total of 3,626 complementary determining region 3 (CDR3) sequences were extracted across samples for comparison. These findings represent an important crucial step towards the application of immune sequencing for clinical monitoring and risk stratification of ARDS patients.

BACKGROUND

- ARDS is an acute disorder that begins within 7 days of the inciting event. It is characterized by infiltrates in both lungs and severe progressive hypoxemia, as well as damage to the capillary endothelium and alveolar epithelium.
- Endothelial damage leads to the movement of fluid and macromolecules into interstitial and pulmonary air spaces, causing pulmonary edema.
- T-cells are implicated as playing a role in the modulation of the aberrant response leading to excessive tissue damage and, eventually, ARDS.
- Most of the diversity in T-cell receptors (TCRs) is contained in the CDR3 regions of the heterodimeric cell-surface receptors.

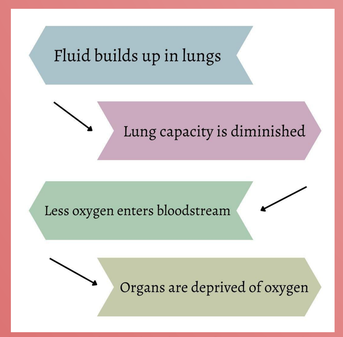

Figure 1: The proposed mechanism of ARDS pathogenesis and the downstream physiological effects.

METHODS

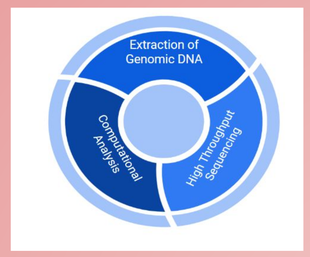

Figure 2a

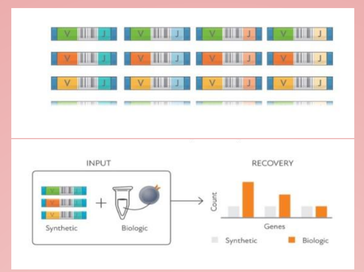

Figure 2b

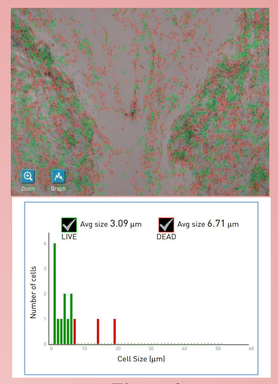

Figure 2c

Figure 2a: The experimental workflow employed in the study.
Figure 2b: Overview of the ImmunoSEQ software. immunoSEQ is a multiplex PCR-based assay that uses high-throughput sequencing to measure the CDR3 variable chain of T-cell and B-cell receptors. This allows for the characterization of the diversity of the immune repertoire.
Figure 2c: This figure illustrates how cell viability is measured and captured visually. Prior to extracting genomic DNA from samples, this method was used to assess them. The Countess instrument, an Artificial Intelligence device, was employed to quantify biospecimens for this study. It detects fluorescence-tagged cells using laser beams, capturing the light absorbed by these cells. In the top portion of the figure, the cell regions outlined in green indicate the presence of live cells while cell regions outlined in red indicate the presence of dead cells. The lower portion of the figure shows the quantitative analysis that accompanied the visual representation.

RESULTS

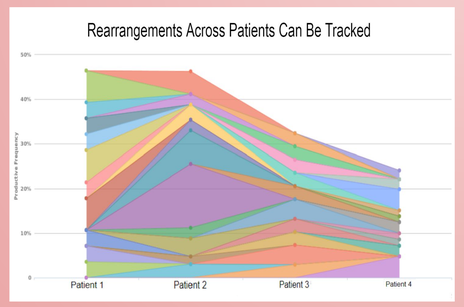

Figure 3a

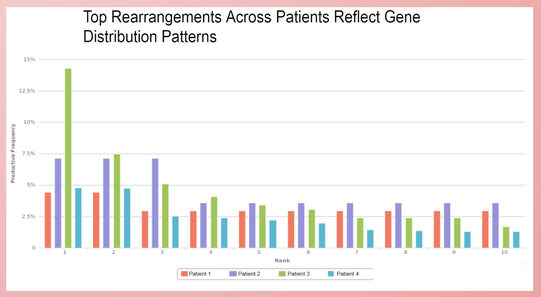

Figure 3b

3D Models of CDR3 Amino Acid Sequences Derived from Top Gene Rearrangements

Figure 4a

Figure 4b

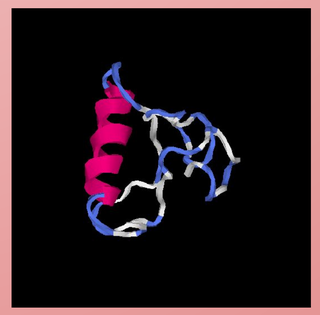

Figure 4c

Figures 3a: VDJ Gene Rearrangements shown for representative samples from ARDS patients. The top gene rearrangements which are shared by the samples are shown based on productive frequency.
Figure 3b: Ranking of the gene rearrangements is shown for each of the same representative samples displayed in figure 3a. Genes are ranked according to productive frequency.
Figure 4a-c: CDR3 sequences derived from gene rearrangements were extracted and modelled using the iTASSER platform. Images corresponding to respective top rearrangement were extracted from sequence reads. Classification of the highest scoring models enabled discernment of biophysicochemical features and deeper analysis of the structural relationships.

RESULTS

Immunogenicity scores obtained from CDR3 sequences

Sample ID	Positive Scoring Sequences	Total Number of Sequences	Percent % Immunogenicity Observed	Clinical Diagnosis
111	515	1560	33.01%	ARDS
176	19	56	33.93%	ARDS
186	501	1485	33.74%	ARDS
251	64	202	31.68%	ARDS
158	7	25	28.00%	Non-ARDS
162	33	118	27.97%	Non-ARDS
163	51	169	30.18%	Non-ARDS

Table 1: Immunogenicity scores in the table were retrieved from CDR3 sequences for representative samples. For samples from representative ARDS patients, the percentage of CDR3 sequences with positive immunogenicity scores ranges from ~32–34%. For the non-ARDS samples, the percentage is slightly lower in the range of ~28–30%. The difference in the variance of immunogenicity scores between the ARDS samples and non-ARDS samples was statistically significant ($p < 0.05$).

CONCLUSIONS

- CDR3 sequences from lung edema fluid exhibit distinct clonal populations.
- CDR3 sequences can be further characterized based on biochemical features.
- Analysis of these CDR3 sequences offers insight into the CDR3-driven T-cell repertoire of ARDS.
- The data obtained from this study can be applied to monitor specific T-cell populations in ARDS patients, enhancing our comprehension of how the adaptive immune system contributes to disease pathogenesis.

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